

Metrology

Science of Measurement

- Meteorology - Science of weather forecast.
- Metallurgy - Science of extraction of metals from ores.

Types of Metrology

- (i) **Fundamental/Scientific metrology** which focuses on the development and management of measurements.
- (ii) **Industrial/Workshop Metrology** which ensures the accuracy and functionality of measurement instruments
- (iii) **Medical Metrology** to ensure and improve the traceability of measurement quantities in the field of health.
- (iv) **Legal Metrology** which oversees the precision of measurements in economic transactions.

The main objective of Metrology is to establish and maintain the required accuracy of measurements at the minimum possible cost. This involves ensuring the reliability of measuring instruments, standardizing measurement methods, and minimizing errors to achieve consistent and accurate results in various applications.

Error is the deviation desired value(difference between measured value and desired/nominal value. Errors are bound to occur in nature.. Nothing or nobody is 100% perfect in this world.

Error can be minimized or reduced depending on requirements under given situations.

Process is a set of interrelated activities or interacting elements which transforms inputs into useful outputs.

Inputs of a process – (i) Man and Management (ii) Machines (iii) Materials(iv) Methods and Measurement method (v) Milieu(Operating conditions or Environment)

Manufacturing Process is a set of operations/activities that transforms raw materials or components into finished goods.

Requirement is need or expectation that is stated, generally implied or obligatory.

Any product may have many properties or distinguishing features or characteristics with upper specification limits(USL) and/or lower specification limits(LSL) with or without target(T).

There are four possibilities in Engineering and applied sciences :only LSL, only USL, LSL-USL & LSL-T-USL.

e.g. Heart beat of human body lies between 72 beats per minute(LSL) and 80 beats per minute(USL)

Blood sugar :**70 - 110 mg/dl**, Temperature: **37°C +/- 1°C** etc. SpO₂ : **90 - 100 %**

Specification is a document stating requirements of an object(product)

Inspection - Determination of conformity to specified requirements (by measurement and/or testing and/or gauging accompanied by judgment)

Quality is Degree to which a set of inherent characteristics of an object fulfills requirements. (ISO 9000:2015)

Indian Standard 1 (IS 1) has been revised multiple times. Initially published in 1951, it was revised in 1964 and then again in 1968 as IS 1:1968. The 1968 revision, titled "**Specification for the National Flag of India (Cotton Khadi)**", is a bilingual (English and Hindi) document. It outlines the design, construction details, and other specifications for the Indian National Flag made from hand-spun and hand-woven cotton khadi bunting.

IS 2347 - Specification of Pressure Cooker, IS 14543 - Specification of Packaged drinking water,

IS - 366 Specification of Electric Iron(Calender), IS/ISO 9000/ISO 9001/ISO 9004 - Specification of Quality Management Systems

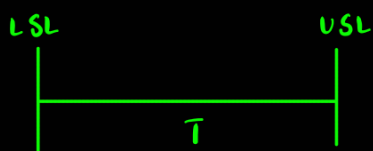
There are more than 20000 Indian Standards as on date.

- ① It is science of measurement.
- ② Similar names - meteorology, metallurgy.
(weather) (minerals)
- ③ Types -

- (i) Fundamental (V.C. scale)
- (ii) Industrial (Industry)
- (iii) Legal (Diff. in measuring units)
- (iv) Medical (pulse-rate meters, etc.)

Standard specifications -

- | | | |
|--------------|---------------|--------------|
| (i) IS 1 | (iii) IS 366 | (v) IS 10645 |
| (ii) IS 2347 | (iv) IS 14543 | |



$LSL \rightarrow$ Lower specification limit
 $USL \rightarrow$ Upper specification limit
 $T \rightarrow$ Target

$LSL, USL, LSL - T - USL, LSL - USL$

Eg. - sugar cub. - $LSL - USL = (70 - 110) \text{ mg / deci-l}$

Some definitions

Errors: Deviation from desired value.

Specification: It is a document stating all req. of a prod., or a sys., or a process.

IS is formulated by BIS in New Delhi.

(Indian standards)

BS	EN	MILSTD	GOST	JIS
↓	↓	↓	↓	↓
British	Europe	USA (military)	Russia	Japan (standards)

IS 1: specification of Indian Flag (colours, cloth, dimension, Ashoka chakra).

IS 2347: specification of pressure cooker

IS 366: Iron press

IS 14543: Portable drinking water (packaged drinking water)

IS 10645 : spec. of a process

ISI → Indian Standard Institute

ISO 9001:2015 : req. for quality management systems (QMS).
↓
International Standards
[Plan-do-check-Act] cycle

- ① Creation of new item / prod. / process / object.
 - ② Process is a set of inter-related activities or interactive elements which convert inputs into output. (IS 105)
 - ③ Output of a process is object (product).
 - ④ Input of a process -
 - ① man / management
 - ② machines
 - ③ materials
 - ④ methods
 - ⑤ milieu (operating condition or environment)
- } def. by IS & ISO

Eg- $\phi 50 \pm 0.02 \text{ mm}$ ⇒ $USL = 50.02 \text{ mm}$
 ↙ Target ↘ errors $LSL = 49.98 \text{ mm}$
 dia. of a curve (say shaft) $T = 50 \text{ mm}$

① Deviation from desired values.

② It is of 2 types -

- (i) Tracable / controllable errors
- (ii) Untracable / uncontrollable errors

③ Sources -

- (i) Instrumental errors
- (ii) Human errors
- (iii) Environmental errors
- (iv) Operational errors
- (v) Loading / unloading errors

$$\sigma_{\bar{n}} = \sqrt{\frac{\sum_{i=1}^n (n_i - \bar{n})^2}{n-1}}$$

== Precision $\neq 0$

$\bar{n}$ = Avg. mean
 n_i = readings
Not n as it will increase the error
so we will go for more control if its
 n then it is rms of S.D.

$$\sigma_T = \sqrt{\frac{\sum_{i=1}^n (n_i - T)^2}{n-1}}$$

== Accuracy $\neq 0$

T = True value

Inspection: Conformity evaluation as per specification requirements.

Quality: It is a degree to which a set of inherent (permanent) characteristics of object which fulfill its requirements.

Degree: Rank or category of an element.

Conformity: Lies between USL & LSL, also called conformance.

All prod. must be in limits otherwise, it is useless.

Accuracy: Quantitative measure of amount of closeness of measured value to the true value (T).

Precision: Quantitative measure of amount of closeness of measured value to the mean value ($\bar{n}$).

Slip gauge is a rectangular block & is used to check if micrometer is giving correct readings or not. [can give screw gauge's accuracy]

Repeatability: other name of precision as it gives repeatability of values measured by instrument.

Gauge: gives reading b/w USL & LSL.

Calibration: A process of repairing a faulty measuring, testing or gauging instrument in order to bring it to a desired level of performance by comparing the same with same level or higher level measuring, testing or gauging instrument.

$$\text{Standard Deviation } (\sigma) = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

$$\text{Accuracy} = \sigma_T = \sqrt{\frac{\sum_{i=1}^n (x_i - T)^2}{n-1}}$$

$$\text{Precision} = \sigma_{\bar{x}} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

Given True value = $T = 100 \text{ mm}$

From observation/readings $\bar{x} =$

$$\frac{100.001 + 99.997 + 100.002 + 99.998 + 100.003}{5}$$

$$= \frac{500.001}{5} = 100.0002 \text{ mm}$$

$$\text{Then } \sigma_T = \sqrt{\frac{(100.001-100)^2 + (99.997-100)^2 + (100.002-100)^2 + (99.998-100)^2 + (100.003-100)^2}{4}}$$

$$= 0.00259807 \text{ mm}$$

$$\text{and } \sigma_{\bar{x}} = \sqrt{\frac{(100.001-100.0002)^2 + (99.997-100.0002)^2 + (100.002-100.0002)^2 + (99.998-100.0002)^2 + (100.003-100.0002)^2}{4}}$$

$$= 0.002588435 \text{ mm}$$

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